PYTHON PRACTICE WORKSHEET

Scenario-Based Coding Challenges | Beginner Level

Topics: Python basics, data types, variables, operators, if-else, and loops

How to use this worksheet: Read each real-life situation, identify the inputs and logic, then write and test a Python program. Use meaningful variable names and add comments where helpful.

Part 1 — Python Basics, Data Types & Variables

1. Student Profile Card
A school wants to store a student's details. Create variables for the student's name, age, class, percentage, and whether the student is enrolled. Choose a suitable Python data type for each value, then print the details in a neat format.

2. Movie Ticket Bill
A cinema stores the movie name, ticket price, number of tickets, and whether the customer has a membership. Create variables using suitable data types. Calculate and display the total ticket cost.

3. Smart Water Bottle
A smart bottle records how much water a person drinks in litres. Store the person's name, daily water target, and amount already consumed. Print the remaining amount. Decide which values should be integers, floats, or strings.

4. Online Shopping Cart
A customer adds 3 notebooks costing ■45.50 each and 2 pens costing ■10 each. Store item names, prices, and quantities. Calculate the total bill and print its data type using type().

5. Temperature Converter
A weather app receives a temperature in Celsius as a decimal number. Convert it to Fahrenheit using F = (C × 9/5) + 32. Display the result rounded to two decimal places.

Part 2 — Operators in Real-Life Situations

6. Splitting a Restaurant Bill
Four friends share a restaurant bill of ■1,860 equally. Calculate how much each person must pay. Also calculate the amount left if they each pay ■500.

7. Mobile Data Usage
A phone plan includes 2 GB of data. A user has consumed 1.35 GB. Use arithmetic operators to calculate the remaining data. Use a comparison operator to check whether more than 0.5 GB is left.

8. Discount Calculator
An online store gives a 15% discount on a product priced at ■2,400. Calculate the discount amount and final price using arithmetic operators.

9. Event Entry Check
A museum allows entry to visitors aged 12 or above. Store a visitor's age and use comparison operators to check whether the visitor meets the age requirement.

10. Login Validation
A website asks for a username and password. Store the entered username and password, then use logical operators (and/or) to check whether both match the expected credentials. Print an appropriate message.

Part 3 — If, Elif & Else Decision Making

11. ATM Withdrawal
An ATM asks the user for a withdrawal amount. If the amount is greater than the available balance, show 'Insufficient balance'. Otherwise, if the amount is a multiple of 100, allow the withdrawal and update the balance. In other cases, ask the user to enter an amount in multiples of 100.

12. Student Grade Calculator
Ask for a student's marks out of 100. Display the grade using these rules: 90–100 = A, 75–89 = B, 60–74 = C, 40–59 = D, below 40 = Fail. Also handle marks outside the 0–100 range.

13. Ride Eligibility
An amusement park ride requires a visitor to be at least 120 cm tall and at least 10 years old. Ask for age and height. Use if-else and logical operators to decide whether the visitor can ride.

14. Electricity Bill
A household's electricity bill is calculated using units consumed: up to 100 units = ■3 per unit; 101–200 units = ■5 per unit; above 200 units = ■8 per unit. Ask for units and calculate the bill using if-elif-else.

15. Weather Advice App
Ask the user for the current temperature. If it is below 15°C, suggest wearing a jacket. From 15°C to 29°C, suggest comfortable clothing. At 30°C or above, suggest staying hydrated and wearing light clothes.

Part 4 — Loops: For & While

16. Daily Study Tracker
Ask the student how many hours they studied each day for 7 days. Use a loop to collect the hours, calculate the total study time, and display the average per day.

17. Multiplication Table Generator
Ask the user for a number and print its multiplication table from 1 to 10 using a for loop.

18. Savings Goal
A student saves ■50 every day. Use a while loop to show the running savings until the total reaches or exceeds ■1,000. Display how many days it takes.

19. Password Attempts
A system gives a user 3 chances to enter the correct PIN. Use a loop to keep asking until the PIN is correct or all attempts are used. Display whether access is granted or locked.

20. Canteen Order Counter
A canteen wants to record the price of each item ordered. Ask how many items the customer is buying, then use a loop to input each price and calculate the final bill.

Part 5 — Mixed-Concept Mini Projects

21. Quiz Game
Create a 5-question quiz using input(), if-else, and a loop. Award 1 point for each correct answer. At the end, display the score and a message based on the score.

22. Number Guessing Game
Choose a secret number in your code. Give the player up to 5 attempts to guess it. After each guess, tell them whether the secret number is higher or lower. Stop when they guess correctly or run out of attempts.

23. Personal Expense Tracker
Ask how many expenses the user wants to enter. Use a loop to collect each expense and calculate the total. Then use if-else to display 'Within budget' if the total is ■2,000 or less, otherwise display 'Budget exceeded'.

24. Library Fine Calculator
Ask how many days late a library book is. No fine is charged for 0 days. Charge ■2 per day for the first 5 late

days and ■5 per day for each additional day. Calculate and display the fine.

25. Food Delivery Bill
Ask the user to choose a food item and quantity. Calculate the subtotal, apply free delivery for orders of ■500 or more (otherwise add ■40), and print the final bill. Use variables, operators, if-else, and suitable data types.

BONUS CHALLENGE

26. Mini Banking System: Build a menu-driven program that repeatedly displays: 1) Check balance, 2) Deposit money, 3) Withdraw money, 4) Exit. Use a while loop to keep the menu running, if-elif-else to process the selection, and operators to update the balance. Do not allow a withdrawal greater than the available balance.

Teacher tip: Encourage learners to solve the questions independently first. Ask them to explain their logic before running the code. For extra practice, test boundary values such as 0, negative numbers, and values just above or below a limit.